

MAC 2312 — CALCULUS II

Lecture 2: Area Between Curves

Wednesday, August 26 • Sections 2.1

• Guided Notes •

How to Use These Guided Notes: These guided notes are designed to help you organize the main ideas from each lecture but

1. we will not complete every example in class.
2. Although most definitions and theorems are included, you may need to add additional notes, examples, or explanations during class.
3. Occasionally, you will see sections labeled **Deeper Insight**. These sections provide interesting background information and help develop a deeper understanding of calculus, but they are not part of the core Calculus II content.

Today's lecture is devoted to one application of integration: the area of planar regions.

Deeper Insight

A recurring theme throughout this course will be

Integrating lengths produces areas.

A vertical or horizontal segment has a length. By accumulating infinitely many lengths, we obtain an area.

Next week, we will see that

Integrating areas produces volumes.

Understanding what quantity is being accumulated is often more important than memorizing a formula.

Before calculating the area of a region, we must be able to describe and graph the region.

Calculating areas integrating with respect to x

Definition 2.1 – Area Between Two Curves

If f and g are continuous on $[a, b]$ with $f(x) \geq g(x)$ for all x in $[a, b]$, then the area A of the region bounded above by $y = f(x)$, below by $y = g(x)$, and on the sides by $x = a$, $x = b$ is

$$A = \int_a^b [\text{_____} - \text{_____}] dx.$$

Understanding the definition What does the integrand $f(x) - g(x)$ represent geometrically?

It represents the _____ of a vertical segment.

Example 2.1 – Area of the region enclosed by the graphs of two functions

Determine the area of the region enclosed by the graphs of the following functions

$$f(x) = x^2 \text{ and } g(x) = 2x.$$

Algebraic work:

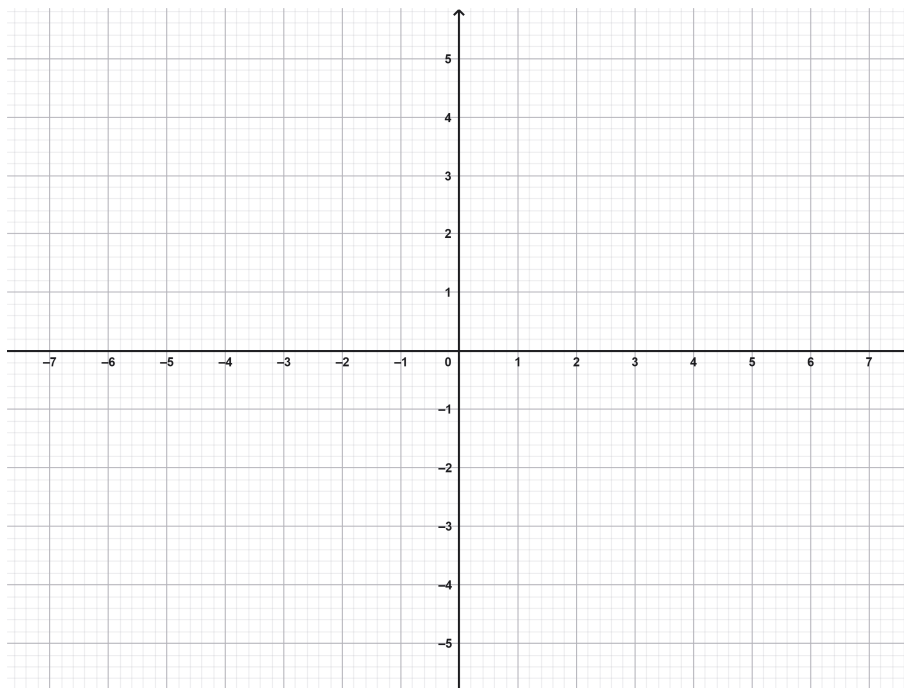
Quantity	Value
Determine the x -coordinates of the intersection points (sort them in ascending order):	
Test points (one value between every pair of consecutive numbers above):	
$T(x)$: top function for every pair of consecutive values above	
$B(x)$: bottom function for every pair of consecutive values above:	

Set up definite integral: in this example, there are only two intersection points

$$\text{Area} = \int_0^{\boxed{}} \text{_____}$$

Evaluate definite integral:

Graphical verification (optional, but recommended):



One last comment: The area of the region enclosed by the graphs of the given curves is obtained by integrating the lengths of the vertical segments extending from the bottom curve to the top curve.

Let us do the same with other regions.

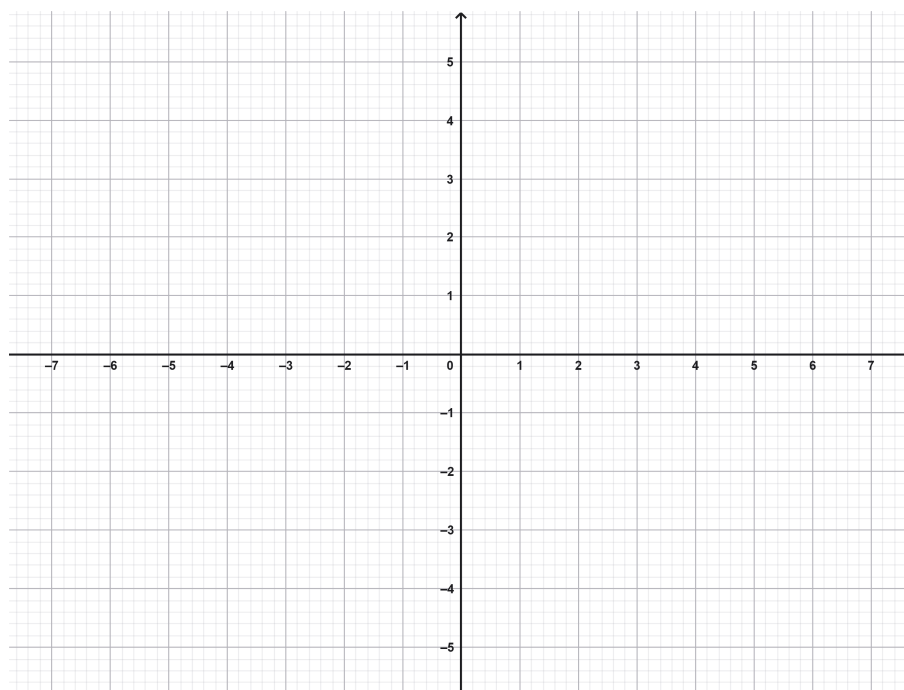
Example 2.2 – Area of the region enclosed by the graphs of two functions between two vertical lines

Find the area of the region enclosed by the graphs of $f(x) = x^2 - 2x - 3$ and $g(x) = x - 3$ over $[-1, 1]$ (equivalent to find the area of the region enclosed by the graphs of the functions and the vertical lines $x = -1$, $x = 1$).

Algebraic work:

Quantity	Value
Left boundary, x -coordinates of any intersection points within the interval $[-1, 1]$ (sorted in increasing order), right boundary:	-1 , _____, 1 .
Test points (one value in each interval):	
$T(x)$:	
$B(x)$:	

Graphical verification:



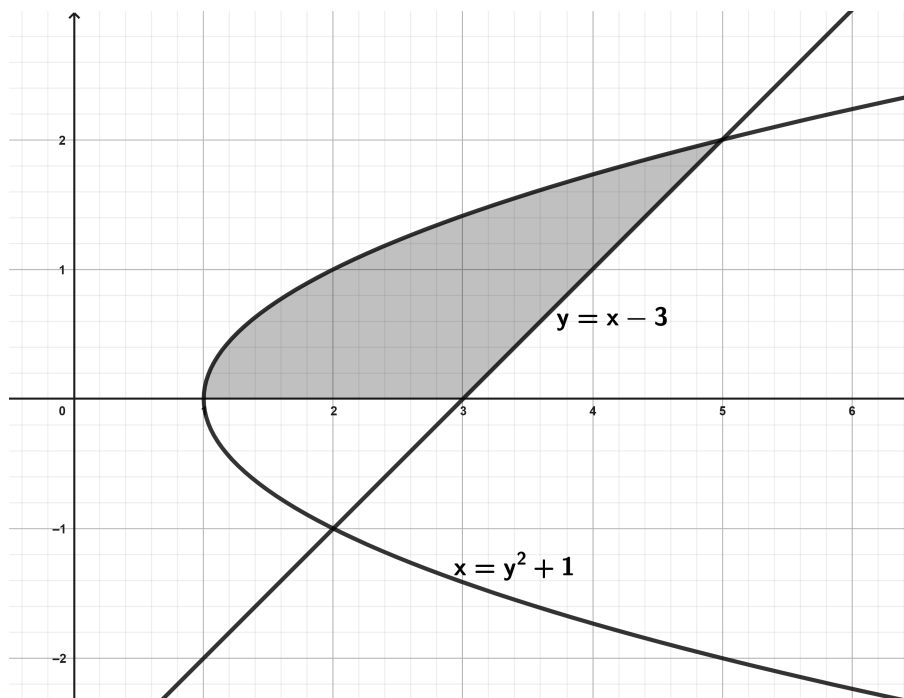
Set up definite integral: in this example, we need to evaluate two definite integrals

$$\text{Area} = \int_{-1}^{\boxed{}} \text{_____} + \int_{\boxed{}}^1 \text{_____}$$

Calculate the area:

Example 2.3 – Area of a region given graphically

Find the area of the shaded region in the graph below.



Reading the graph:

Quantity	Value
Left boundary, x -coordinate of intersection points between boundaries (sorted), right boundary:	
$T(x)$, top function for every pair of consecutive points above	
$B(x)$, bottom function for every pair of consecutive points above:	

Set up definite integral: in this example, we need to evaluate two definite integrals

Area = _____

Calculate the area:

Calculating areas integrating with respect to y

Let us change our perspective.

When integrating with respect to x :

$$\text{Area} = \int_a^b (\text{top}(x) - \text{bottom}(x)) \, dx.$$

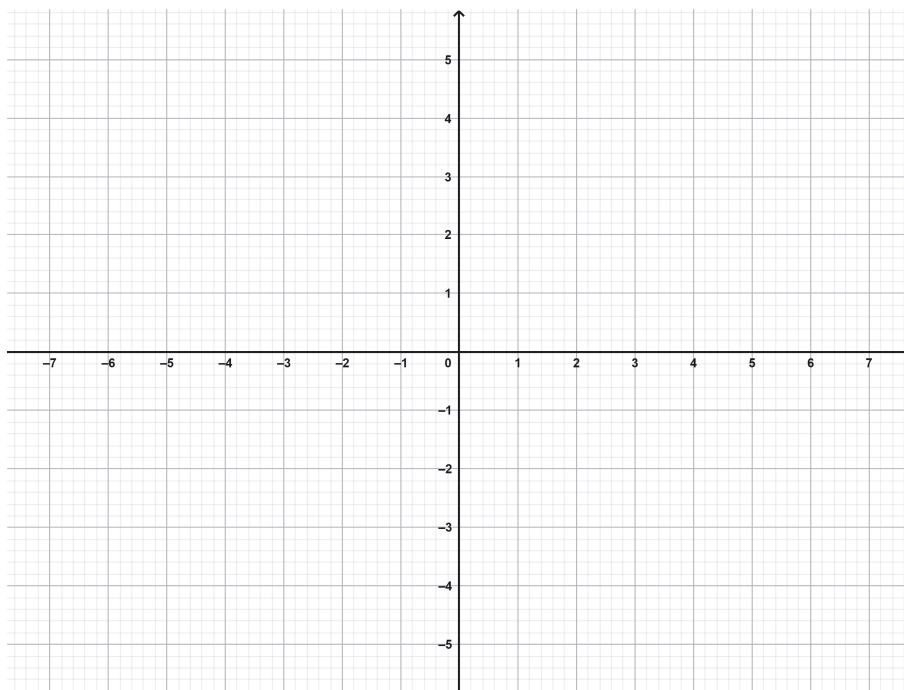
Now, we want to integrate with respect to y :

$$\text{Area} = \int_c^d (\text{right}(y) - \text{left}(y)) \, dy.$$

In both cases, the integrand represents the length of a vertical or horizontal segment in the region.

Example 2.4 – Area of the region enclosed by the graphs of two functions

Find the area of the region enclosed by the graph of the parabola $y^2 = 2x + 6$ and the line $y = x - 1$.



Let us repeat the same steps as in the previous examples, keeping in mind that we now want to integrate with respect to y . Instead of finding the x -coordinates of the intersection points, we need the y -coordinates. Likewise, instead of identifying the top and bottom functions, we must identify the left and right functions.

Quantity	Value
y -coordinates of intersection points:	
Test points (one value in each interval):	
$R(y)$:	
$L(y)$:	

Set up the integral:

Calculate the area:

And now, try it yourself, remember that:

1. A picture is worth a thousand words: when in doubt, sketch the region.

Integrating with respect to x :

- (a) determine x -coordinates of the leftmost and rightmost points.
- (b) Are there any intersection points in between? Which are their x -coordinates?
- (c) Sort all the x -values in increasing order.
- (d) Between every pair of consecutive values in the list above: which is the top function? which is the bottom function? All functions must be written as functions of x .
- (e) Set up the integrals.
- (f) Calculate the area.

Integrating with respect to y :

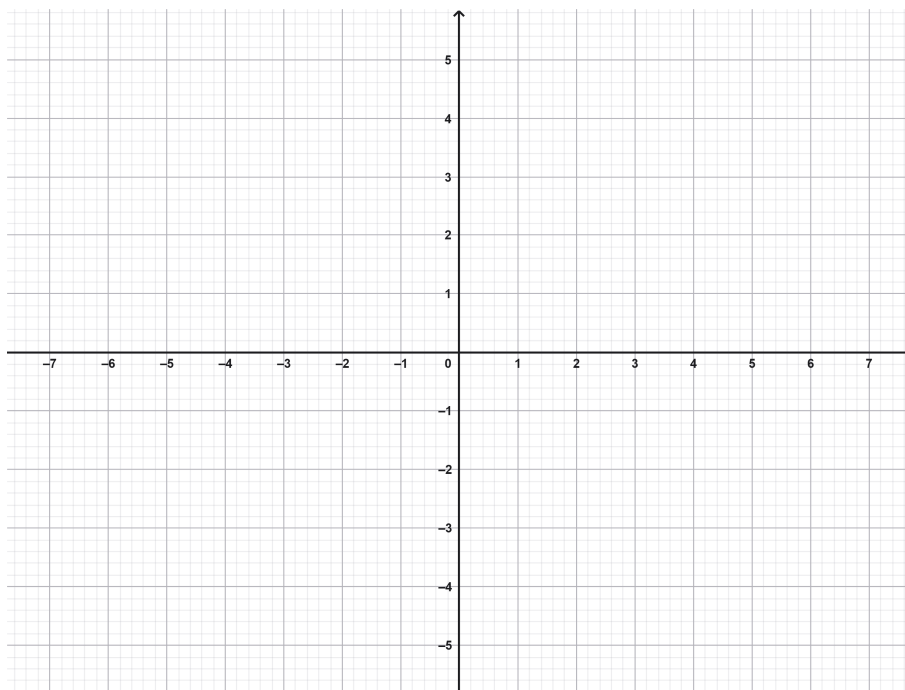
- (a) determine y -coordinates of lowest and highest points.
- (b) Are there any intersection points in between? Which are their y -coordinates?
- (c) Sort all y -values in increasing order.
- (d) Between every pair of consecutive values in the list above: which is the right function? which is the left function? All functions must be written as functions of y .
- (e) Set up the integrals.
- (f) Calculate the area.

2. Usually, we do not specify whether to integrate with respect to x or y . You should choose the variable of integration that leads to the simplest setup.

! Common Mistakes

1. Skipping the sketch: a picture reveals the correct limits and functions.
2. Forgetting to find all intersection points.
3. Assuming the same function remains on top (or on the right) throughout the entire region.
4. Mixing dx and dy : writing dx with an integrand that depends on y or viceversa, integrating with respect to y , but the integrand depends on x .
5. Accepting a negative area without checking the setup.

Try it yourself. Let R be the region enclosed by $y = x$ and $y = x^3$ for $-1 \leq x \leq 1$. Sketch R , find where the curves cross, and set up *and evaluate* the integral for the area of R .

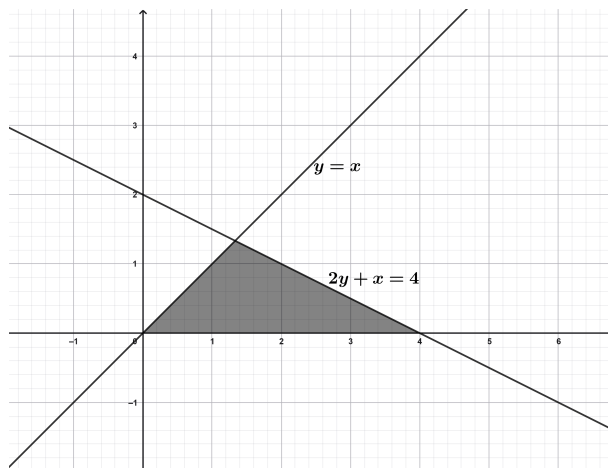


Practice Problems

Level I problems

1. Find the area of the region enclosed by the graphs of $f(x) = x^2$ and $g(x) = x + 2$.
2. Let R be the region enclosed by the parabolas $y = x^2$ and $y = 2x - x^2$.
 - (a) Sketch both curves on the same axes and shade R .
 - (b) Find all intersection points algebraically.
 - (c) Set up and evaluate an integral for the area of R .

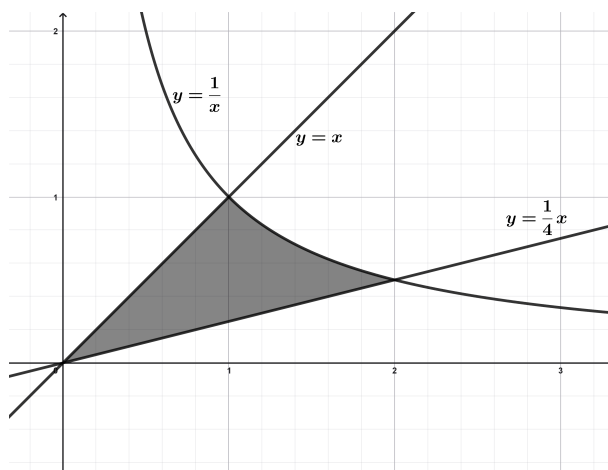
3. Find the area of the region enclosed by $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.
4. Determine the area of the region bounded above by $y = e^x$, bounded below by $y = x$, and on the sides by $x = 0$ and $x = 1$.
5. Find the area of the shaded region in the graph below.



Practice Problems

Level II problems

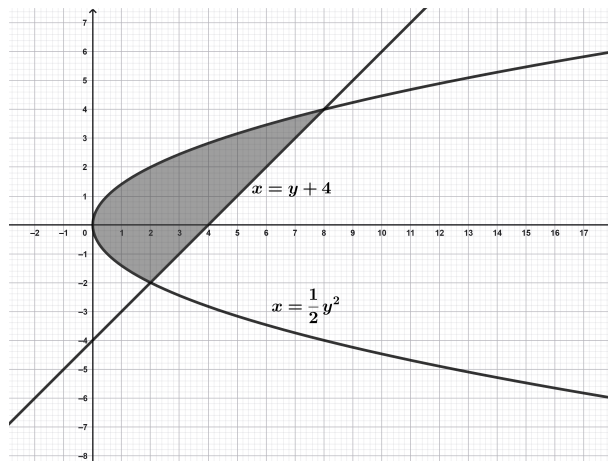
1. Find the area between $y = \sin x$ and $y = \cos x$ on $[0, \frac{\pi}{2}]$. (Hint: where do the curves cross on this interval?)
2. Find the area of the region enclosed by the parabola $y^2 = 2x + 6$ and the y -axis.
3. Find the area of the region bounded above by the graph of $f(x) = x + 3$ and below by the graph of $g(x) = 2|x|$.
4. Find the area of the shaded region in the graph below.



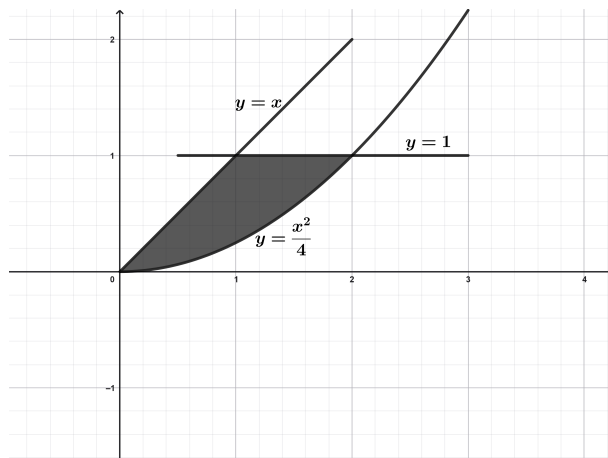
Practice Problems

Level III problems

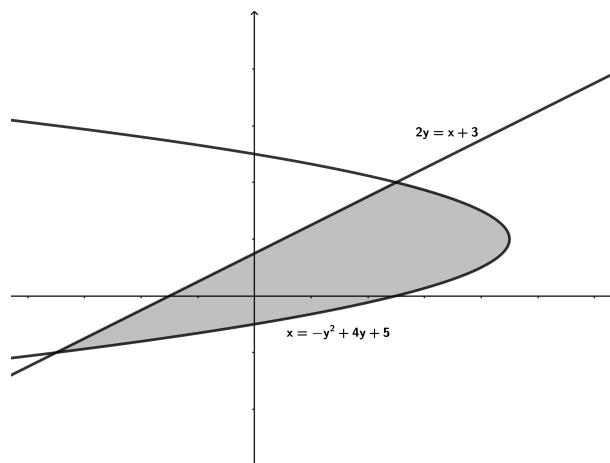
1. Find the area of the shaded region.



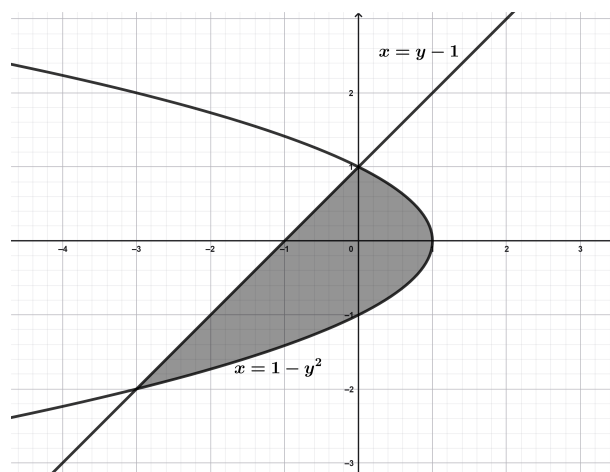
2. Find the area of the shaded region.



3. Find the area of the shaded region.



4. Find the area of the shaded region.



Checking Your Understanding

1. Suppose you set up $\int_{-1}^1 (x - x^3) dx$ for the region bounded by $y = x$ and $y = x^3$ on $[-1, 1]$, without splitting the interval, and get exactly 0. Why should an area of zero feel wrong here? What did the computation actually measure instead of area?
 2. In your own words, describe a step-by-step strategy for setting up *any* area-between-curves problem, starting with “sketch the region.”
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After this lesson I can:

- Sketch and describe a region enclosed by curves.
- Determine where two curves intersect and use those points to define limits of integration.
- Decide whether integrating with respect to x or y is more efficient.
- Identify the top and bottom functions (or right and left functions) needed to set up an area integral.
- Set up and evaluate the definite integrals that represent the area of a planar region.
- Calculate the area of a region enclosed by curves, lines, and/or coordinate axes.

Muddiest Point: What concept from today's lesson is still unclear?